

E-Commerce Customer Intelligence & Revenue Optimization Report

1. Executive Summary

This report presents an in-depth analysis of e-commerce transactional data to identify key revenue drivers and customer behavior patterns.

The study leverages SQL and Power BI to generate actionable insights focused on customer segmentation, retention, and revenue optimization.

Key finding: A small segment of high-value customers contributes a disproportionately large share of total revenue.

2. Objectives

- Analyze revenue trends over time
- Understand customer purchasing behavior
- Identify high-value and at-risk customers
- Provide data-driven recommendations for business growth

3. Data & Tools Used

Dataset: 100K+ transactional records (orders, customers, pricing)

Tools: SQL (PostgreSQL), Power BI

SQL Techniques: Joins, Aggregations, Window Functions (RANK, NTILE)

Visualization: Interactive dashboards with KPIs and filters

4. Methodology

- Data Cleaning & Transformation using SQL
- Aggregation of customer-level metrics (frequency, monetary value)
- RFM Segmentation to classify customers
- Pareto Analysis to study revenue concentration
- Visualization of insights using Power BI dashboards

5. Key Analysis & Findings

Revenue Trends: Strong growth observed, especially between 2017–2018

Customer Segmentation: High-value and loyal customers identified using RFM scoring

Behavioral Analysis: Scatter plots used to evaluate frequency vs spending

Pareto Insight: Top ~20% customers contribute ~70–80% of revenue

6. Business Insights

- Revenue is highly concentrated among a small group of customers
- High-value customers show consistent and frequent purchasing behavior
- A large portion of customers fall into low-engagement segments (potential churn risk)
- Customer retention is a critical factor for revenue sustainability

7. Recommendations

- Implement targeted retention strategies for high-value customers
- Introduce loyalty programs to increase repeat purchases
- Re-engage at-risk customers through personalized offers
- Focus marketing efforts on high-return customer segments

8. Dashboard Overview (Power BI)

The dashboard includes:

- KPI cards (Revenue, Customers, Orders, AOV)
- Revenue trend analysis
- Customer segmentation visuals
- Pareto analysis chart

9. Conclusion

The project demonstrates how data-driven analysis can uncover critical business insights. Combining SQL-based data processing with Power BI visualization enables effective decision-making and strategic planning.